Lab 02

Cleaning water and air quality data

Teaching Week 02 | Individual work. Submit in R or Python.

Purpose

Environmental measurements may contain missing values, inconsistent units, and results below a reporting limit. Build an ordered cleaning script, preserve the original values, and show how your decisions change the summaries used to answer one descriptive question.

Learning outcomes

After completing this lab, you should be able to

- diagnose type, unit, missingness, and reporting-limit problems;
- separate raw data from reproducible transformations;
- compare distributions before and after cleaning; and
- justify rather than conceal exclusions and imputations.

Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

Option A | Surface Water Quality Monitoring Network Grab Sample Water Quality Data

Publisher. Nova Scotia Environment and Climate Change
Official record. https://data.novascotia.ca/d/wncu-ppda
Record identifier. `wncu-ppda`
Licence and format. Nova Scotia Open Government Licence; CSV
Expected variables. station; sample date; parameter; result; unit
Use in this lab. clean and transform water-quality measurements
Why it fits. Long-form measurements support checks of missing values, units, and reporting limits.

Filter to a manageable station, period, and set of parameters before exporting CSV. The record includes `CharacteristicName`, `ResultValue`, `ResultUnit`, detection conditions, and quantitation limits; the spelling of exported headings may differ. Read those fields together. One laboratory result is one row, so several rows may describe the same water sample. Do not average different chemicals or incompatible units together.

Option B | National Air Pollution Surveillance (NAPS) Program

Publisher. Environment and Climate Change Canada
Official record. https://open.canada.ca/data/en/dataset/1b36a356-defd-4813-acea-47bc3abd859b
Record identifier. `1b36a356-defd-4813-acea-47bc3abd859b`
Licence and format. Open Government Licence – Canada; CSV query or ZIP archive
Expected variables. station; date; pollutant; concentration; unit
Use in this lab. clean and compare air-quality observations
Why it fits. Station records support type repair, missingness analysis, and transformation.

Follow the record's Data Query Tool or ECCC Data Catalogue link. Select a station, pollutant, period, and sampling interval; retrieve the corresponding data dictionary and quality-code definitions. Keep one sampling interval in each comparison, and distinguish missing codes from measured zeroes.

Option C | Canadian Air and Precipitation Monitoring Network (CAPMoN)

Publisher. Environment and Climate Change Canada
Official record. https://open.canada.ca/data/en/dataset/4baa2ee4-a8aa-457a-af26-aa13e96ee2f4
Record identifier. `4baa2ee4-a8aa-457a-af26-aa13e96ee2f4`
Licence and format. Open Government Licence – Canada; ECCC Data Mart with delimited data files
Expected variables. site; date; chemical; concentration; precipitation
Use in this lab. prepare atmospheric and precipitation measurements
Why it fits. Repeated chemistry records support checks of units, missingness, and table shape.

Follow the record's Data Mart link and choose one measurement product, site, and period. Read its format and quality-code documentation before importing. Use the delimiter stated for that product; document any conversion to CSV. Confirm that your extract meets the observation requirement before committing to the option.

Data contract

Choose one listed dataset and construct a cleaned observation-level table.

- Retain at least 200 observation-level records and three variables used in your question, including at least one measured quantity. Identifiers alone do not satisfy the analytical-variable requirement.
- Include at least one variable that needs missing-value, type, unit, or reporting-limit treatment.
- Identify any station, site, region, or time columns needed to interpret missingness.
- Record every transformation and preserve the raw value in the reproducible workflow.
- Preserve detection limits and quality flags. A result below a reporting limit is not an observed zero, and an unrecognized token is not automatically a missing value.
- Compare summaries before and after cleaning.
- Reuse Week 01 summaries and unusual-value checks.

- Document the official record URL, publisher, licence, retrieval steps, and access date.

What the contract does not settle

Meeting the contract does not establish that a transformation is scientifically justified.

Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I identified the observational unit, groups, units, identifiers, and missing-value codes.
- ☐ My table satisfies every rule in the data contract.
- ☐ I identified at least one consequential cleaning or transformation decision.
- ☐ I will compare summaries before and after cleaning.
- ☐ I will preserve the raw values and document every transformation.
- ☐ I will cite the dataset in the report.

Cumulative method

This lab carries forward Week 01 observational-unit definitions, summaries, spread measures, and unusual-value checks. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

Work program

1. Profile the raw data

Reproduce the Week 01 variable audit and summaries before changing any value.

2. Specify the cleaning contract

Define valid types, units, ranges, missing codes, duplicate rules, and the observational unit required by your question. Repeated station-date pairs are not necessarily duplicates; parameter, depth, sample fraction, and laboratory identifiers may distinguish them. Use the publisher's documentation to decide, and count repeated keys as shown in Class B before removing any row.

3. Implement transformations

Write one ordered script that preserves raw inputs and records counts affected by each decision. Use the Class B cleaning example as a pattern. Document any type repair or unit conversion, including the conversion formula and source. Reshape only if your question needs a different table layout.

4. Test the consequences

Compare non-missing counts, missingness, and the Week 01 summaries before and after cleaning, within comparable measurement units. Reuse the histogram or boxplot taught in Week 01 for one consequential change. Investigate whether exclusions are concentrated by site, time, or another meaningful group. Keep censored results distinguishable and explain which summaries exclude them.

5. Report limitations

Explain one cleaning choice that could change the conclusion and one additional fact you would need to resolve it. No imputation, sampling, PCA, predictive model, or significance test is required; use a method only when it answers your question and you can justify its assumptions.

Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Downloaded data and generated results are not course source, so they never go into the course repository. Commit the report, its source, and the generated tables and figures it cites to your private work repository. Keep that evidence private until the deadline, and publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-02/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on September 24, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-02/`. A branch URL or movable tag alone is not a fixed submission. The package includes

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, figures, and any summary statistics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

Assessment standard

The course uses one rubric across all integrated labs. The best eight of the nine weekly lab results contribute 15% of the course grade.

Table 1. Integrated lab marking criteria.

Criterion	Weight (%)
Dataset provenance and suitability	15
Reuse of earlier methods	20
Current-week analytical method	35
Interpretation and limitations	20
Reproducibility and communication	10
Total	100

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.